

ELEVATOR SYSTEM LLD – INTERVIEW NOTES

1. Explain overall architecture...

- Floor -> ExternalButton -> Dispatcher -> Scheduler -> Strategy -> Controller -> PQ -> Thread wake.

2. Why two PQs?

- LOOK algorithm requires directional batching: min-heap for UP, max-heap for DOWN.

3. Concurrency issues:

- PQ race conditions.
- movingDirection/nextFloorStoppage races.
- currentFloor visibility.
- Missed notify/wait.
- Stop flag atomicity.
- Lifetime issues.

4. Simultaneous button presses:

- Independent flows.
- Scheduler is read-only safe.
- Mutex protects PQ.
- notify_one wakes thread.
- LOOK ensures ordered processing.

5. Purpose of CV + Mutex:

- Avoid busy wait.
- Exclusive PQ access.
- Efficient wakeup.

6. Why not global queue?

- Breaks strategy.
- No direction locality.
- Starvation.

7. Internal vs External dispatcher:

- External selects elevator.
- Internal routes to same elevator.

... (Truncated for brevity in this PDF example)